What is it about maximally entangled states that is so fascinating? I can sum this up in two statements:

- An entangled state is a complete description of the combined system. No more can be known about it.

- In a maximally entangled state, nothing is known about the individual subsystems.

How can that be? How could we know *as much as can possibly be known* about the Alice-Bob system of two spins, and yet know *nothing* about the individual spins that are its sub-components? That's the mystery of entanglement, and I hope that by the end of this lecture you will understand the rules of the game, even if the deeper nature of entanglement remains a paradox.

6.8 ALICE AND BOB'S OBSERVABLES

So far, we've discussed the space of states of the Alice-Bob two-spin system, but not its observables. Some of these observables are obvious, even if their mathematical representation is not. In particular, using their apparatuses $\mathcal{A}$ and $\mathcal{B}$, Alice and Bob can measure the components of their spins:

$$\sigma_x,\ \sigma_y,\ \sigma_z$$

and

$$\tau_x,\ \tau_y,\ \tau_z.$$

How are these observables represented as Hermitian operators in the composite space of states? The answer is simple. Bob's operators act on Bob's spin states exactly as they would if Alice had never shown up. The same goes for Alice. Let's review how the spin operators act on the states of a single spin. First, let's look at Alice's spin: